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Where Minds Pollinate

Physics
Class 11

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Chapter 7
Handwritten Notes

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CHAPTER: 7

PHYSICS OF SOLID

"The branch of physics that deals with structure and physical properties is called physics of solid."

They have fixed shape volume and incompression

TYPES OF SOLIDS

Crystalline Solids	Amorphus solids
Those solids in which particles (atoms, ion, molecule) are arranged regularly throughout in three dimensional pattern	Those solids in which particles (atoms, ion, molecule) are not arranged in regular manner are called Amorphous solid
Arrangement: Particles are arranged in periodic way	Arrangement: Particles are not arranged in regular manner.
Melting point: They have sharp and fixed melting point.	Melting point: They have not sharp and not fixed melting point.

Amorphus → without shape

ISOTROPIC

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Isotropic and Anisotropic nature

Crystaline solids are anisotropic in nature	Amorphus solids are isotropic in nature

Heat of fusion

They have definite value of heat of fusion.	They do not have definite value of heat of fusion

Nature of Solid

Crystaline solids are true solids	Amorphous solids are pseudo solid or super cooled liquid.

Examples

» ionic crystals (NaCl)(KCl) » Metals (Fe, Cu) » Non-metalic crystals diamond » Semiconductor (Si)(Ge)	Glass, Glue, plastic "At 800°C a glass is converted into paste"

POLYCRYSTALLINE SOLIDS

"Polycrystaline solid is a material made up of many small crystals oriented in random direction"

- These small crystal are called crystalides or grain
- The size of grain is 10^{-3}m – 10^{-6}m.

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These grains are separated by well-defined boundary.

They are isotropic in nature.

They have low specefic gravity.

Polythene $(C_2H_4)_n$, polystyrene $(C_8H_8)_n$, polyvinyl chloride $(C_2H_3Cl)_n$, natual rubber $(C_5H_6)n$.

CRYSTAL AND ITS PROPERIES

Basis:

Basis is a collection of atoms in particular fixed arrangement in space.

Lattice:

The collection of infinite number of points in periodic arrangement is called a lattice.

Space lattice:

It is a skeleton upon which crystal structure is build by placing atoms on or near the lattice point.

lattice sites:

The position of particles in a crystal lattice is called lattice sites.

Lattice constant:

The separation distance between two consective lattice sites is called lattice constant

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Lattice transational vector:

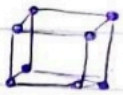
When a lattice constant is assigned a direction it will become lattice transational vector.

Unit cell:

The smallest basic 3-D structure which define all the properties of crystal is called unit cell.

TYPES OF UNIT CELL

1. Simple cubic unit cell:



Atoms used = 1atom

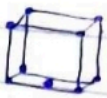
$8 \times 1/8$ = 1unit

8sides = 1atoms

1sides = $1/8$ atoms

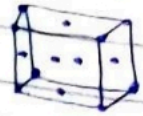
2. Body centered unit cell:

2atoms per unit cell



3. Face centered unit cell:

4 atoms per unit cell



every face = half atom

6 face = 3atoms

Total atom = 3+1 atoms

= 4 atoms

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MECHANICAL PROPERTES OF SOLID

"The properties of solid in response to applied force are called mechanical properties"

Deforming force:

"A force which change length shape or volume of the body is called deforming force."

- In solid deforming force is expressed in term of stress.

Deformation:

"The change in length, shape or volume due to deforming force is called deformation."

Elasticity:

"The ability of body to restore its original length, shape or volume is called elasticity"

Plasticity:

"The ability of body to get permanent deformation is called plasticity."

Ductile substance:

"Such substance which undergoes plastic deformation until they break are called ductile substane."

e.g metals,

- Brittle substance.

In simple cubic crystal 8 atoms at each corner is placed from fi

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such substance which break just after the elastic limit is reached are called brittle substance.

for example, glass, high carbon steel

Stress: (σ)

"The applied force per unit area is called stress"

$$\sigma = \frac{F}{A}$$

Unit: $Nm^{-2} \Rightarrow kgm^{-1}s^{-2}$

TYPES OF STRESS

Tensile stress:

"Such stress which change the length of body is called tensile stress."

Volumentric stress:

"A stress which change the volume of body is called volumentric stress."

Shear stress:

"A stress which change the shape of body is called volu shear stress."

- All stress have same formula and same unit.

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Strain : (ε)

"The measure of deformation is called strain"

ε = change in quantity / original value of quantity

TYPES OF STRAIN

Tensile strain:

"It is the ratio of change in length to the original length."

$$\varepsilon = \frac{\Delta L}{L}$$

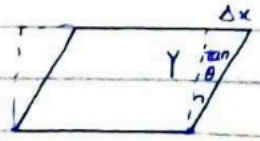
Volumentric strain:

"It is the ratio of change in volume to the original volume."

$$\varepsilon = \frac{\Delta V}{V}$$

Shear strain:

"It is the ratio of displacement of sheared face to the displacement of fixed plan face.



$$\varepsilon = \frac{\Delta x}{Y}$$

all strain have different formula but no unit

$\varepsilon = \tan\theta$

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$\varepsilon = \theta$ (if θ is very small $\tan \simeq \theta$)

Hooke's Law

"The stress applied on a body is directly proportioned to the strain produced in it which is elastic limit is called Hooke's law".

stress ∝ strain

stress = constant strain

$\frac{\text{stress}}{\text{strain}} = \text{constant}$

$\frac{\text{stress}}{\text{strain}} = \text{modulus of elasticity}$

Modulus of Elasticity

"It is the ratio of stress of the strain with the elastic limit is called modulus of elasticity"

$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\varepsilon}$

SI unit: Nm^{-2}

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TYPES OF MODULUS OF ELASTICITY

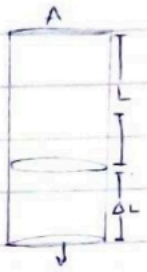
Youngs Modulus (Y)

"It is the ratio of tensile stress to the tensile strain is Young's modulus.

Formula: ∛ $Y = \frac{\text{Tensile stress}}{\text{Tensile strain}}$

$$Y = \frac{F/A}{\Delta L/L}$$

$$Y = \frac{F \times L}{A \times \Delta L}$$



Unit: Nm^{-2}

Dependence: modulus of material

Bulk- modulus: (B)

"It is the ratio of volumentric stress to the volumentric strain"

Formula:

$$B = \frac{\text{Volumentric stress}}{\text{Volumentric strain}}$$

$$B = \frac{F/A}{\Delta V/V}$$

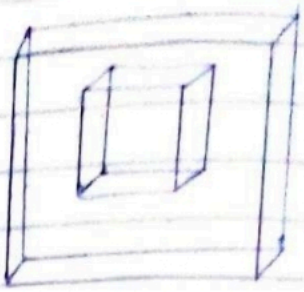
$$B = \frac{F \times V}{A \times \Delta V}$$

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SI unit: N/m^2

formula:

Compressibility $\propto \frac{1}{\text{bulk modulus}}$



Shear modulus:

"It is the ratio of shear stress to shear strain"

formula: $S = \frac{\text{shear stress}}{\text{shear strain}}$

[Gases and liquids only have bulk modulus]

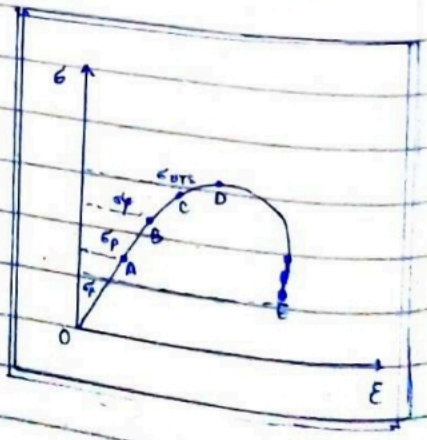
$S = \frac{F/A}{\Delta x/Y}$

$S = \frac{F/A}{\tan\theta} \Rightarrow S = \frac{F/A}{\theta}$

$S = \frac{F}{A\theta}$

⇒ STRESS STRAIN CURVE FOR DUCTILE MATERIAL:

If a tensile metal wire is extended at a specific deformative rate and stress in the wire during deformation curve



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or continuously measured by suitable electronic device.

A typical stress-strain curve for a ductile material.

OA-region:

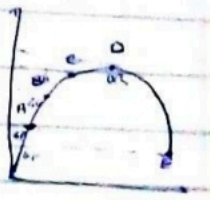
Hooke's law is valid at OA region.
Stress-strain has linear relation.
Body restore its original dimension when applied force is removed.
Solid behaves as an elastic body.

Proportional limit σ_p:

The maximum value of stress within the elastic limit is called proportional limit.

AB-region:

Hooke's law is not valid at AB region.
Stress-strain has no linear relation.
Body restores its og dimension when force is removed.
Point B is called "elastic point" and corresponding stress is called elastic limit."



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Elastic limit:

"The maximum value of stress after that permanent deformation will start is called elastic limit."

→ **B-C region:**

- Material undergoes permanent deformation
- Body does not recover its original shape and dimension.
- The region is called "plastic region".
- point C is called yield point.
- and corresponding stress is called yield stress.

Yield stress:

The value of stress at which material will begin to deform plastically is called yield stress.

→ **CD region:**

- Material becomes permanently deformed
- Body does not recover its shape and dimension.
- This region is called "plastic region".
- strain rapidly increases even for small change in stress.
- Point D of the curve show "ultimate tensile stress"

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ULTIMATE TENSILE STRESS (σUTS)

"The maximum value of stress that material can with stand is called ultimate tensile stress."

DE region:

"It's stress is zero but strain is not zero this type of deformation is called "plastic deformation"

Point E on the curve is called "breaking point" and corresponding stress is called "fracture stress"

stress strain curve for brittle substance



Fracture Stress:

"The value of stress at which material break into pieces"

* Slope of stress - strain curve:

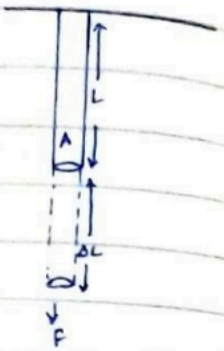
Young's modulus: $\frac{\sigma}{\varepsilon}$

* Area under stress - strain curve:

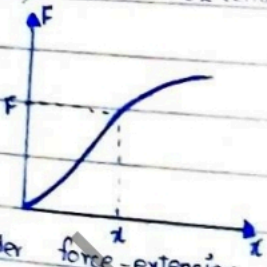
Strain energy density $= \sigma \times \varepsilon$

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WORK-DONE FROM FORCE EXTENSION GRAPH AND STRAIN ENERGY:



Consider a wire of length 'L' and area of cross section 'A' is suspended from rigid support as shown in figure. When weight is suspended and gradually increases, the length of wire also increases and has direct relation within the elastic limit. If graph is plotted between force (on y-axis) and extension (on x-axis)



Area under force-extension graph (within elastic limit)

Work done = Area of triangle

$= \frac{1}{2}(F)(\Delta L)$

$W = \frac{1}{2} F \Delta L$ —— i

Here F = average force

extension = 0 Force = 0

extension = x Force = F

$$W = \frac{1}{2} F \Delta L$$

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Strain-energy:

"The energy stored in deformed material due to their displacement from mean position is called strain energy"

The work done required to deform a material

$$W = \frac{1}{2} F \Delta L \quad \text{—i}$$

According to Hooke's law

Force $\propto$ extension

$$F \propto \Delta L$$

$$F = K \Delta L$$

putting value of F in above equation

$$W = \frac{1}{2} (K \Delta L) \Delta L$$

$$W = \frac{1}{2} K \Delta L^2$$

This work done is stored as strain energy or elastic potential energy

Potential energy = strain energy = $\frac{1}{2} K \Delta L^2$

From Young's modulus of material.

$$Y = \frac{F \times L}{A \times \Delta L}$$

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From here $F = \frac{YA\Delta L}{L}$

putting value of F in eq i

$$W = \frac{1}{2}\left(\frac{YA\Delta L}{L}\right)\Delta L$$

$$W = \frac{1}{2} YA \frac{\Delta L^2}{L} \quad -ii$$

Strain-energy density:

strain-energy per unit volume is called strain-energy density.

$$\text{strain-energy density} = \frac{\frac{1}{2} F \Delta L}{A L}$$

$$= \frac{1}{2}(F/A)(\Delta L/L)$$

$$\text{strain-energy density} = \frac{1}{2} \sigma \times \varepsilon \quad -i$$

* $\sigma \times \varepsilon = 2$(strain energy density)

$$\text{strain energy density} = \frac{1}{2} Y \varepsilon^2 \quad -ii$$

$$= \frac{1}{2} \frac{\sigma^2}{Y} \quad -iii$$

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